

Female Reservation and the Qualifications of Local Elected Officials in India

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Working draft

Abstract

How does reserving an office for women change the qualifications of the person elected? We examine rural panchayats in Bihar, Uttar Pradesh, Rajasthan and Kerala, and urban councillors in Mumbai. Education is measured consistently by the share with a university degree. Among rural village heads, the adjusted reserved-minus-open difference is -15.0 percentage points (95% CI -16.6 to -13.5) in Bihar, and negative in each UP election. Rajasthan's classified-winner estimate is negative, but missing education prevents an all-winner conclusion. Kerala's village ward members show smaller, imprecise differences. In Mumbai, reserved-seat councillors have a higher estimated graduate share, with an interval that includes zero, and fewer pending criminal cases. These comparisons control for geography and, in the rural analyses, caste reservation. Their causal interpretation requires conditional comparability of reserved and open seats. The evidence concerns qualifications at election; it does not establish how reservation changes performance in office.

1 Introduction

Reserving elected offices for women changes who can contest them. Does it also change the qualifications of the people elected? The relevant comparison is between winners of women-reserved and open seats. Open seats can elect women, so a comparison of female and male politicians does not answer this question.

Existing rural studies often find lower schooling and less political experience under reservation. Their evidence spans different offices, outcomes and research designs. Some compare candidates rather than winners; others compare women in reserved seats with men in open seats. These distinctions determine which findings bear on the reservation effect on elected elites.

This paper brings that evidence together with administrative records of elected officials. The rural analyses cover village heads and council members in Bihar, Uttar Pradesh, Rajasthan and Kerala; the urban analysis covers Mumbai councillors. Graduate share provides a common education measure. The comparisons remain separate by office and election, and rural panchayats remain separate from urban councils.

Education, experience and pending criminal cases describe different attributes. Their value as measures of elite quality depends on their relationship to the work of governing. A degree records formal education, while a pending case records an allegation. Neither alone establishes competence or performance. Age describes the composition of elected officials and is reported separately.

2 Existing evidence

Lower schooling and less experience under reservation appear in several rural studies. In the 2001 working paper’s West Bengal sample, reserved council heads have 7.13 rather than 9.92 years of schooling, a difference of -2.79 years (SE 0.54); 11% rather than 43% had previously been elected to the council (Chattopadhyay and Duflo 2001, Table 3). In the subsequent Birbhum council term, the adjusted schooling gap is -2.10 years (SE 0.55), and reserved leaders are 26.5 percentage points more likely to be first-time pradhans (SE 6.8) (Beaman et al. 2009, Table II, column 6). These samples overlap geographically and are not independent national tests.

The May 2007 southern-state draft, published in 2008, reports -2.62 years of schooling (SE 0.77) and a political-knowledge gap of -1.41 points (SE 0.29). That regression excludes caste-reserved councils (Ban and Rao 2007, Table 9). Karnataka’s descriptive comparison reports degrees among none of the 23 reserved female presidents and among 19% of the 48 male presidents. Illiteracy is 9% and 4%, respectively; the comparison also conditions on the leader’s gender, so it is not a clean reservation contrast (Raabe et al. 2009, Tables 3–4).

The 2011 working paper on the NCAER village panel, published in 2015, reports illiteracy of 24.72% in reserved observations versus 5.74% in unreserved observations, secondary education or above of 28.65% versus 62.69%, and previous officeholding of 11.67% versus 24.24%. These are descriptive comparisons across 180 reserved and 459 unreserved observations (Deininger et al. 2011, Table 3). The 2013 Andhra Pradesh working paper, published in 2017, reports that 15% of reserved heads versus 54% of unreserved heads have higher-secondary schooling or more; 11% versus 22% have prior political experience. The characteristics table covers 128 reserved and 169 unreserved heads (Afridi et al. 2013, Table 2).

Recent studies require additional distinctions. The September 2022 Karnataka candidate draft reports lower absolute qualification distributions under reservation (Mori et al. 2022, fig. 2 and Table 3). The published article reports stronger cognitive selection relative to the eligible female population (Mori et al. 2025). That relative benchmark does not establish that reservation increases the elected elite’s absolute qualifications. It also concerns candidates. The March 2026 Bihar threshold draft reports standardized schooling effects of -0.765 SD for candidates (SE 0.024; $N=44,583$) and -0.921 SD for winners (SE 0.073; $N=3,294$). Substituting husbands’ schooling gives -0.147 SD and -0.364 SD, respectively (Bamezai et al. 2026, Table 7). Spousal schooling does not reveal who governs.

Favorable judgments about participation, voter attitudes or policy performance address different outcomes. They can coexist with lower schooling among entrants. Likewise, a positive selection score relative to same-gender citizens can coexist with lower absolute schooling. A synthesis of the reservation effect must keep the outcome, elected status, comparison group and office fixed before combining estimates. The studies reviewed here do not yet supply a harmonized meta-analysis.

3 Data and empirical strategy

3.1 Outcomes and samples

The target is the impact of female reservation on the elected officeholder’s qualification, relative to opening the same office to either sex. An open seat can elect a woman. Comparing female and male winners would answer a different question, as would comparing all candidates rather than the people elected.

The main education outcome is an indicator for a completed university degree or above. Its mean is the graduate share. Illiteracy describes the lower end of the same education distribution and appears alongside graduate share in the appendix. These indicators are mutually exclusive but not exhaustive: a literate non-graduate belongs to neither. Their estimates must not be counted as independent measures of quality or independent studies. Unknown qualifications remain missing.

Separate linear regressions for these binary outcomes estimate changes in their shares; they do not assume that the outcomes are statistically independent. On an identical sample with identical controls, coefficients for a complete set of education categories would sum to zero. Testing a change in the entire education distribution would require a joint test that accounts for the covariance across category estimates. The paper reports pointwise intervals for individual shares, without making that joint claim.

3.2 Specification and inference

For each rural state, election and office, the analysis estimates

$$Y_i = \alpha + \beta R_i + \gamma_{g(i)} + \delta_{c(i)} + \epsilon_i,$$

where R_i records female seat reservation, $\gamma_{g(i)}$ controls for geography and $\delta_{c(i)}$ for caste reservation. Geography is the block when it is populated throughout the relevant slice, and otherwise the district. The district and block together identify a block. Standard errors are clustered at the geographic-control level. All reported intervals are pointwise 95% intervals from `fixest`, with its small-sample variance adjustment and a Student t critical value based on the number of clusters minus one. They do not adjust for testing several offices or outcomes. Kerala’s village and district-member regressions have only 14 district clusters. Their intervals use a few-cluster approximation and warrant particular caution.

For the binary education outcome, 100β is the adjusted reserved-minus-open difference in percentage points. The linear model describes this difference without imposing equal distances between school qualifications. Its causal interpretation requires conditional comparability of reservation assignments and adequate outcome coverage. Geographic and caste controls alone do not verify that requirement. No lottery or discontinuity design is claimed for these administrative-data regressions. Mumbai uses administrative-ward and council effects, with standard errors clustered by electoral ward. That specification does not include caste-reservation controls, so its adjustment set differs from the rural models.

The administrative inputs flow from state election repositories through `local_reservations`. The analysis records input hashes, winner selection, exclusions, outcome classification and regression samples. Bihar’s winners are inferred from unique highest recorded votes or sole uncontested candidacies. UP’s earlier files report elected pradhans; its latest candidate file marks winners. Rajasthan links nomination characteristics to results. Kerala records elected ward members, including those who subsequently hold council leadership positions. The reservation variable for those records refers to their ward, not the presidency.

4 Rural panchayats

4.1 Village heads

Rural village heads: graduate or above

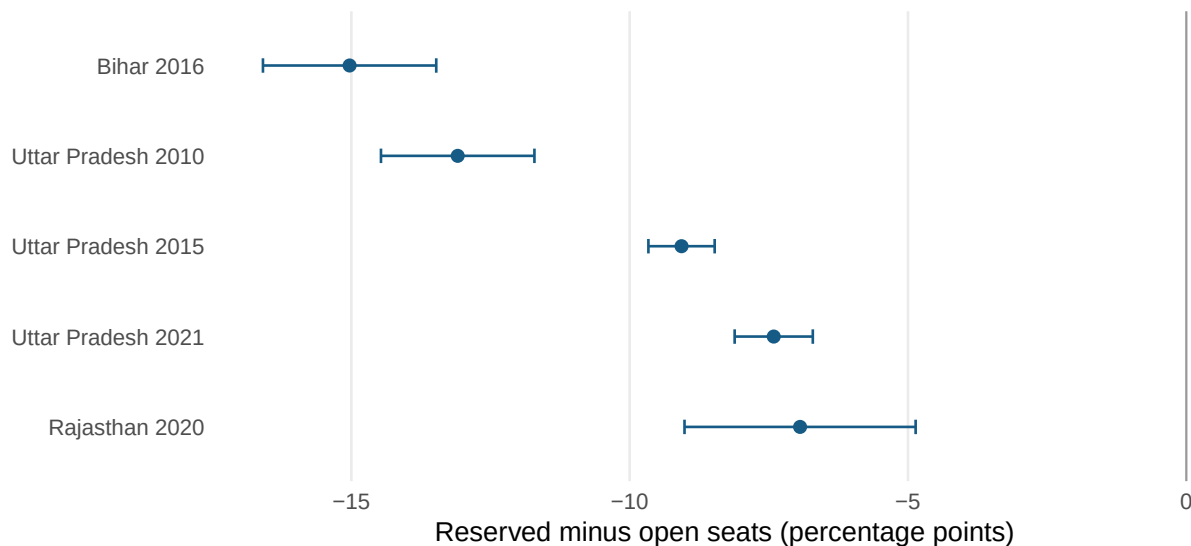


Figure 1: Graduate-share differences among rural village heads. Points compare women-reserved with open seats, conditional on geography and caste reservation; bars are pointwise 95% confidence intervals clustered by geography. UP 2010 uses districts; other rows use blocks. Education is restricted to classified qualifications. Ambiguous winners and unknown reservation are excluded. Sources: state election records through the central election repository.

UP’s graduate-share gap is -13.1 percentage points (95% CI -14.5 to -11.7) in 2010, -9.1 percentage points (95% CI -9.7 to -8.5) in 2015, and -7.4 percentage points (95% CI -8.1 to -6.7) in 2021. These are separate election samples. Changes in their coefficients do not by themselves identify learning, convergence or an effect of repeated reservation.

Rajasthan’s estimate is -6.9 percentage points (95% CI -9.0 to -4.9). Its source includes many qualifications labelled “Other”, which remain unclassified. The estimate therefore describes winners whose graduate status can be classified. Allowing every missing qualification to take either degree status puts the adjusted difference for all eligible winners between -29.7 and 31.5 percentage points. These data alone do not determine its direction.

4.2 Bihar’s other offices

Bihar provides evidence across village, block and district offices. Its analysis includes 210,910 elected seats after excluding ambiguous candidate serials. Unknown education remains missing. Implausible or missing ages do not remove a winner from education regressions.

Bihar 2016: graduate or above

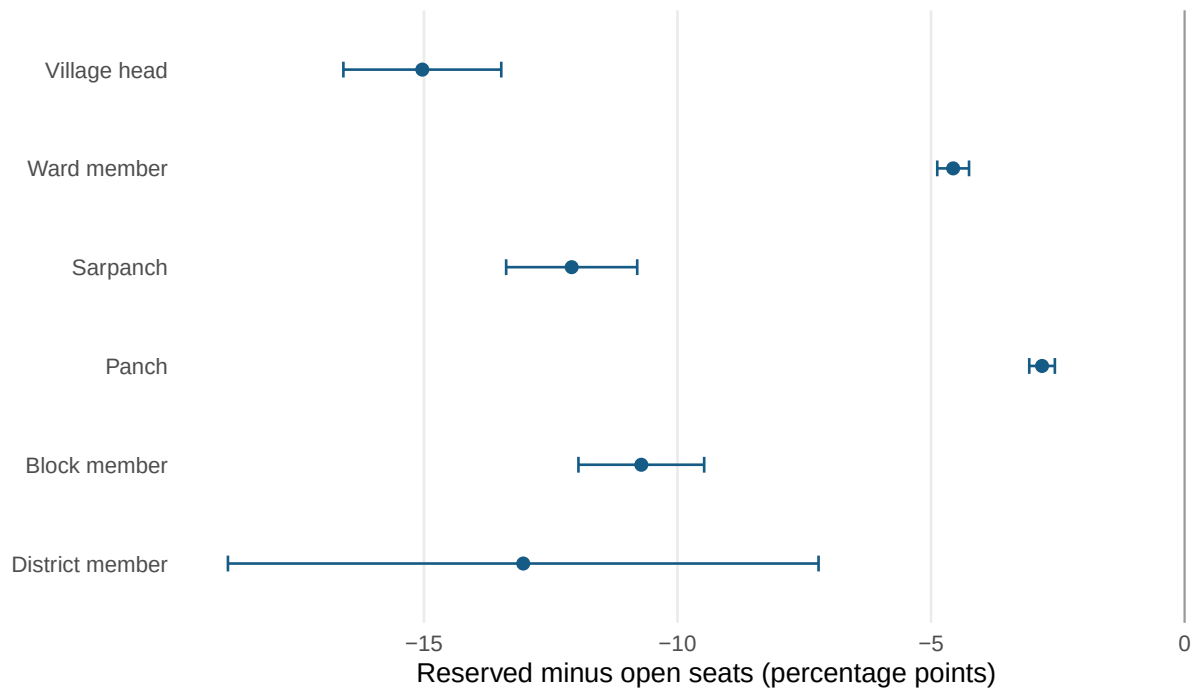


Figure 2: Bihar 2016: graduate-share differences by office. Points are reserved-minus-open differences with block and caste-reservation controls, except district council members, who use districts. Bars are pointwise 95% confidence intervals clustered at that geographic level. Unknown education and ambiguous winner records are excluded. Source: Bihar SEC candidate records through the central election repository.

The graduate-share estimate is negative in every office. The appendix reports illiteracy alongside graduate share as another part of the education distribution. Separate age regressions describe younger winners under reservation.

4.3 Kerala by election and office

Kerala: graduate or above

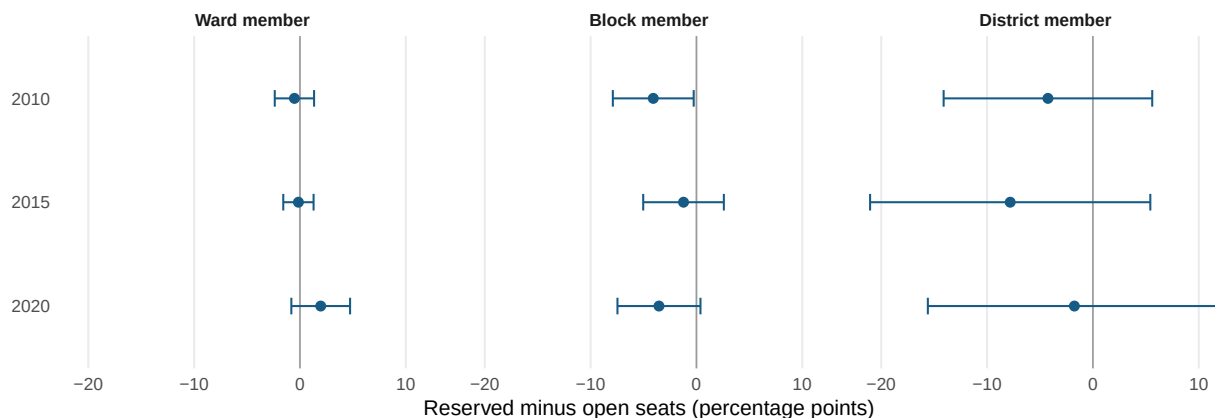


Figure 3: Rural Kerala: graduate-share differences by election and office. Each point comes from a separate regression with caste-reservation controls. Block council members use block effects and clustering; village and district ward members use districts because block identifiers are unavailable. Bars are pointwise 95% confidence intervals. Unclassified qualifications are missing. The common horizontal scale permits comparison across offices. Source: Kerala election records through the central election repository.

For Kerala’s village ward members, the differences are -0.5 percentage points (95% CI -2.4 to 1.3), -0.1 percentage points (95% CI -1.6 to 1.3), and 2.0 percentage points (95% CI -0.8 to 4.7). They provide no precise evidence of the large negative gaps observed among village heads elsewhere. They concern ward members, however, so the contrast with Bihar and UP cannot be attributed to the state alone.

Education is recorded as free text. The coding recognizes reported degrees and school qualifications, leaves incomplete or unclassified qualifications missing, and exports every source label with its classification and frequency. It does not infer that an unmatched string means no degree. These rules differ from the earlier pooled Kerala descriptive table, which used loose substring matches and included unmatched values in its denominator. That historical table is preserved in the repository’s evidence archive rather than treated as an estimate from this specification.

5 Urban councils

5.1 Mumbai

Mumbai’s data link reservation to councillor qualifications in the Praja/affidavit records held in the Karekurve-Ramachandra and Lee replication deposit (Karekurve-Ramachandra and Lee 2023). The sample contains one observation per councillor spell with age recorded. Repeated survey waves do not create additional councillors. The analysis pools the 2012 and 2017 councils with council and administrative-ward effects; standard errors are clustered by electoral ward.

Urban: Mumbai councillors, 2012 and 2017

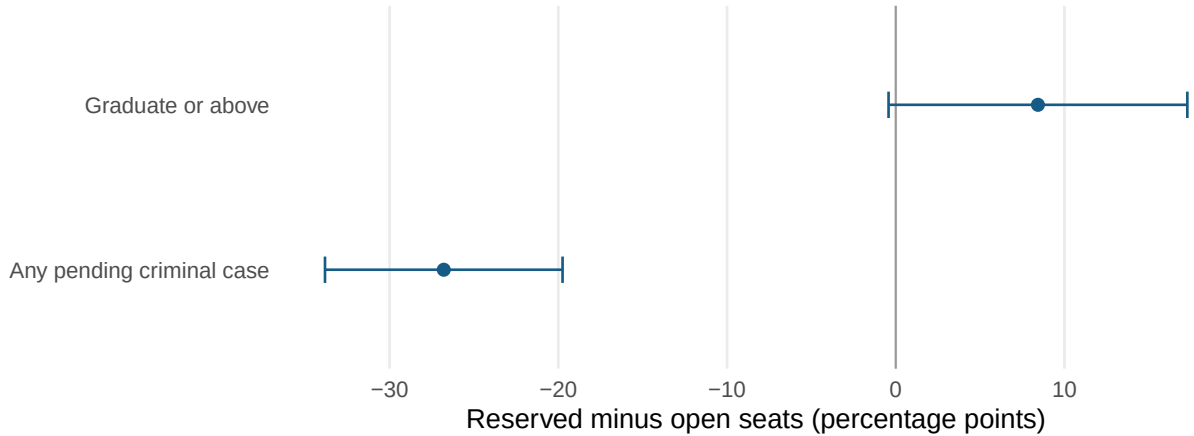


Figure 4: Urban Mumbai: qualifications of elected councillors in the 2012 and 2017 councils. Points are women-reserved-minus-open differences with administrative-ward and council effects; bars are pointwise 95% confidence intervals clustered by electoral ward. One observation is retained per councillor spell with age recorded. Pending cases are allegations, not convictions. Source: Praja/affidavit data in the Karekurve-Ramachandra and Lee replication deposit.

The graduate-share difference is 8.4 percentage points (95% CI -0.4 to 17.3). The pending-case difference is -26.8 percentage points (95% CI -33.8 to -19.7). The education interval permits a small negative effect as well as a positive one; it does not establish that reservation raises education. Pending criminal cases are separately reported because a charge does not establish guilt.

The earlier Rajasthan municipal comparison and Kerala’s urban descriptive tables belong to this urban evidence, not the rural estimates above. Rajasthan’s category “high school or less” includes the twelfth standard; Mumbai’s ends at the tenth. Those levels cannot be pooled as a common outcome. The historical tables remain available in the evidence archive, with their original denominators and qualifications.

6 Discussion

The clearest negative education differences occur among rural village heads in Bihar and UP. Rajasthan’s missing qualifications prevent an all-winner conclusion. Kerala supplies evidence on different offices, with smaller ward-member estimates and wider uncertainty about their signs. Mumbai supplies an urban comparison in which the graduate-share estimate is positive but imprecise, while the pending-case estimate is negative. The paper does not estimate an urban-rural difference or a national average.

A quantitative synthesis would need a common outcome and office, comparable reservation contrasts, and an accounting of shared samples. Schooling years, graduate shares, political knowledge and same-gender selection scores cannot be averaged as though they measured the same quantity. Multiple education thresholds from one sample also do not create independent evidence. The current evidence supports an organized comparison; a pooled reservation-effect estimate requires further harmonization.

7 Appendix: measurement and reproducibility

Table 1: Education coverage among eligible winner records. N includes unclassified qualifications.

State	Year	Office	Seat	Classified (%)	N
UP	2010	Village head	Open	99.6	34,654
UP	2010	Village head	Women	99.4	17,166
UP	2015	Village head	Open	100.0	39,093
UP	2015	Village head	Women	100.0	19,926
UP	2021	Village head	Open	95.3	32,997
UP	2021	Village head	Women	94.1	16,774
Rajasthan	2020	Village head	Open	81.8	5,695
Rajasthan	2020	Village head	Women	56.9	5,433
Kerala	2010	Block member	Open	91.0	983
Kerala	2010	Block member	Women	93.0	1,095
Kerala	2010	Ward member	Open	89.7	8,047
Kerala	2010	Ward member	Women	90.8	8,594
Kerala	2010	District member	Open	87.7	162
Kerala	2010	District member	Women	94.7	169
Kerala	2015	Block member	Open	87.4	1,021
Kerala	2015	Block member	Women	88.7	1,041
Kerala	2015	Ward member	Open	87.6	7,725
Kerala	2015	Ward member	Women	89.7	8,196
Kerala	2015	District member	Open	86.0	164
Kerala	2015	District member	Women	92.2	167
Kerala	2020	Block member	Open	88.7	980
Kerala	2020	Block member	Women	89.3	1,098
Kerala	2020	Ward member	Open	83.8	7,726
Kerala	2020	Ward member	Women	84.9	8,226
Kerala	2020	District member	Open	86.1	166
Kerala	2020	District member	Women	84.2	165

The confidence intervals reflect regression uncertainty conditional on the extracted and classified data. They do not include transcription error, uncertainty about free-text qualifications, or selection from missing education. Rajasthan’s unspecified qualifications and Kerala’s differential classification rates limit claims about all elected officials. Bihar’s age range is restricted to 21–100, as are the new rural age analyses; the education samples retain those with missing age.

The scripts retain source identifiers, sample counts, recoding dictionaries and typed Parquet analysis files. Input hashes pin the local files used here. Some UP inputs come from integration work whose central adapter has uncommitted changes; the repository preserves the adapter diff and byte hashes rather than presenting that state as a published release. The final data release should reconcile those changes upstream. The included analysis snapshots allow inspection of the fitted samples in the meantime.

The 2021 Bihar candidate release lacks statewide structured winner education. Its current reservation feeds and historical source checks are retained upstream; no 2021 Bihar reservation-effect

estimate enters this paper. Delhi’s municipal files are a separate potential urban extension whose source provenance and field coding still need verification.

The missing-education bounds hold the all-winner regression design fixed and allow each unclassified graduate indicator to be either zero or one. After residualizing reservation on the controls, the coefficient is a weighted sum of outcomes. Assigning missing outcomes according to the signs of those weights gives the exact minimum and maximum coefficient. These bounds concern the adjusted coefficient, not the causal effect, and contain no sampling uncertainty. UP’s bounds remain negative in each election; Rajasthan’s and Kerala’s include both signs.

Table 2: Missing-education bounds on the adjusted all-winner graduate-share difference (percentage points). These are not confidence intervals.

State	Year	Office	Lower	Upper	Missing	N
UP	2010	Village head	-13.5	-12.4	254	51,820
UP	2015	Village head	-9.1	-9.1	0	59,019
UP	2021	Village head	-12.0	-1.3	2,543	49,771
Rajasthan	2020	Village head	-29.7	31.5	3,380	11,128
Kerala	2010	Block member	-12.3	3.4	165	2,078
Kerala	2010	Ward member	-10.6	8.9	1,617	16,641
Kerala	2010	District member	-12.7	4.8	29	331
Kerala	2015	Block member	-13.8	10.1	247	2,062
Kerala	2015	Ward member	-12.2	10.5	1,800	15,921
Kerala	2015	District member	-18.0	3.7	36	331
Kerala	2020	Block member	-14.5	7.5	228	2,078
Kerala	2020	Ward member	-14.4	16.9	2,495	15,952
Kerala	2020	District member	-15.1	15.2	49	331

8 Appendix: complete rural estimates

8.1 Bihar: education

Table 3: Bihar rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2016	Village head	-15.0 [-16.6, -13.5]	1.5 [0.9, 2.1]	7,908	523	Block
2016	Ward member	-4.6 [-4.9, -4.3]	11.6 [10.8, 12.3]	97,278	517	Block
2016	Sarpanch	-12.1 [-13.4, -10.8]	4.6 [3.7, 5.6]	7,864	521	Block
2016	Panch	-2.8 [-3.1, -2.6]	14.8 [13.9, 15.7]	85,385	516	Block
2016	Block member	-10.7 [-12.0, -9.5]	3.2 [2.5, 3.9]	10,769	519	Block
2016	District member	-13.0 [-18.9, -7.2]	1.6 [-0.1, 3.4]	889	38	District

8.2 Uttar Pradesh: education

Table 4: Uttar Pradesh rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2010	Village head	-13.1 [-14.5, -11.7]	6.3 [5.1, 7.4]	51,566	72	District
2015	Village head	-9.1 [-9.7, -8.5]	9.3 [8.7, 10.0]	59,019	823	Block
2021	Village head	-7.4 [-8.1, -6.7]	8.2 [7.5, 9.0]	47,228	728	Block

8.3 Rajasthan: education

Table 5: Rajasthan rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2020	Village head	-6.9 [-9.0, -4.9]	0.1 [-0.1, 0.2]	7,748	352	Block

8.4 Kerala: education

Table 6: Kerala rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included. Illiteracy is unavailable.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2010	Ward member	-0.5 [-2.4, 1.3]	—	15,024	14	District
2010	Block member	-4.1 [-7.9, -0.3]	—	1,913	150	Block
2010	District member	-4.3 [-14.1, 5.6]	—	302	14	District
2015	Ward member	-0.1 [-1.6, 1.3]	—	14,121	14	District
2015	Block member	-1.2 [-5.0, 2.6]	—	1,815	148	Block
2015	District member	-7.8 [-21.1, 5.4]	—	295	14	District
2020	Ward member	2.0 [-0.8, 4.7]	—	13,457	14	District
2020	Block member	-3.5 [-7.5, 0.4]	—	1,850	149	Block
2020	District member	-1.8 [-15.6, 12.1]	—	282	14	District

8.5 Age

Table 7: Age of rural officials: differences in years with pointwise 95% confidence intervals. N: winners with age recorded; G: geographic clusters. Controls match the education models.

State	Year	Office	Difference [95% CI]	N	G
Bihar	2016	Village head	-3.6 [-4.1, -3.1]	7,969	524
Bihar	2016	Ward member	-2.0 [-2.2, -1.8]	97,289	517
Bihar	2016	Sarpanch	-5.7 [-6.2, -5.2]	7,894	521
Bihar	2016	Panch	-2.9 [-3.1, -2.7]	85,406	516

State	Year	Office	Difference [95% CI]	N	G
Bihar	2016	Block member	-2.7 [-3.1, -2.3]	10,801	519
Bihar	2016	District member	-4.1 [-5.6, -2.6]	892	38
Uttar Pradesh	2010	Village head	-1.4 [-1.6, -1.2]	51,662	72
Uttar Pradesh	2021	Village head	-1.3 [-1.5, -1.1]	49,757	728
Rajasthan	2020	Village head	-1.7 [-2.3, -1.1]	11,113	352
Kerala	2010	Ward member	-4.8 [-5.9, -3.8]	14,793	14
Kerala	2010	Block member	-4.9 [-5.8, -4.0]	1,980	151
Kerala	2010	District member	-4.2 [-7.3, -1.2]	330	14
Kerala	2015	Ward member	-4.2 [-5.2, -3.2]	15,724	14
Kerala	2015	Block member	-3.9 [-4.9, -3.0]	2,029	151
Kerala	2015	District member	-3.0 [-5.8, -0.3]	327	14
Kerala	2020	Ward member	-3.5 [-4.5, -2.4]	15,860	14
Kerala	2020	Block member	-3.1 [-4.0, -2.2]	2,070	152
Kerala	2020	District member	-3.6 [-6.0, -1.2]	331	14

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